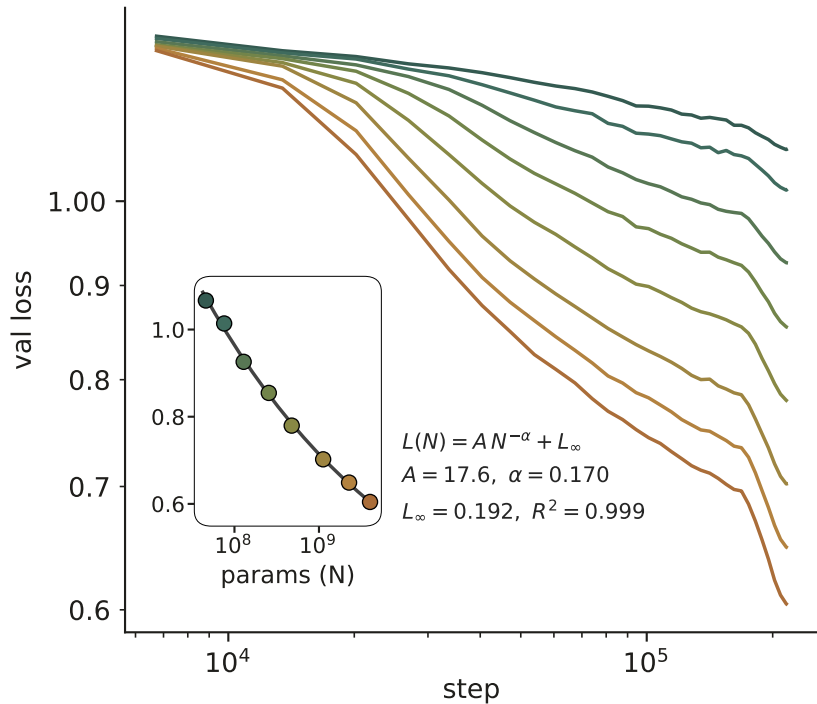
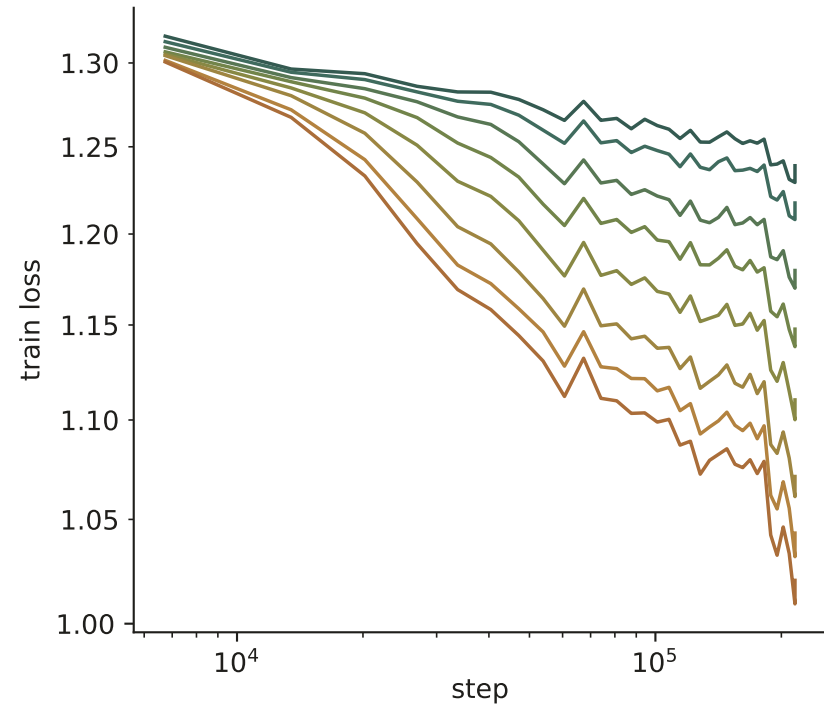


Parameter scaling — loss curves & scaling law

$N = 46\text{M} - 4\text{B}$, $D = 84\text{B}$, $C = 2.5 \times 10^{19} - 2.1 \times 10^{21}$



model params

Model Size	Color
46M	Dark Green
76M	Medium Green
128M	Light Green
255M	Yellow-Green
476M	Yellow
1B	Orange
2B	Brown
4B	Dark Brown